

Guillain-Barré syndrome after COVID-19 vaccination: A systematic review and analysis of case reports

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ABSTRACT

Introduction: Cases of Guillain-Barré Syndrome (GBS) have been believed to be associated with the novel COVID-19 infection, and also with the following vaccines developed against the infection. Our work aims to investigate the incidence of GBS after COVID-19 vaccination, and describe its clinical characteristics and potential confounders.

Methods: An electronic search was conducted through four databases: PubMed, Scopus, medRxiv, and Google Scholar for all case reports and case series describing after COVID-19 vaccine administration. All published articles from inception until November 1st, 2022 were included. Differences between groups were assessed using Pearson chi-square test. Modified Erasmus GBS Outcome Score (mEGOS) for the ability to walk after GBS was calculated for all cases with sufficient clinical data, and Kaplan-Meier survival analysis was performed to study the effect of vaccine type on the relationship between vaccination time and complication of GBS.

Results: About 103 studies describing 175 cases of GBS following COVID-19 vaccination were included. The Acute Inflammatory Demyelinating Polyradiculoneuropathy subtype was the most reported subtype with 74 cases (42.29%). The affected age group averaged around 53.59 ± 18.83 years, with AMSAN occurring in a rather older group (63.88 ± 20.87 years, $p=0.049$). The AstraZeneca vaccine was associated with AIDP ($n=38$, 21.71%) more than other vaccines, $p=0.02$. The bilateral facial palsy subtype was mostly linked to adenoviral vector vaccinations, accounting for an average of 72% of the total BFP cases. Dysesthesias was the most reported sensory complication (60%, $p=0.349$). Most GBS patients survived (96%, $p=0.036$), however, most patients had low mEGOS scores (4 ± 3.57 , $p<0.01$). On average, patients developed GBS at 13.43 ± 11.45 days from vaccination ($p=0.73$), and survival analysis for complication of GBS into mechanical ventilation or walking impairment yielded a severely increased probability of complication after 25 days ($p<0.01$). Intravenous immunoglobulins ($p=0.03$) along with rehabilitation ($p=0.19$) were the most commonly used treatment.

Conclusion: This work investigates the incidence of Guillain-Barré Syndrome after COVID-19 vaccination. Most cases occurred after receiving the AstraZeneca or Pfizer vaccines, and despite low mortality rates, ambulation was compromised in most patients. A higher risk of GBS complication is associated with an onset later than 12–13 days, particularly with Pfizer, AstraZeneca, and Moderna vaccines. No specific predisposing or prognostic factor was identified, and the relation between the COVID-19 vaccines and GBS remain unclear.

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1. Introduction

Guillain-Barré Syndrome (GBS) is a serious immune-mediated neuropathy that occurs rarely post-infection. About 100,000 new cases are diagnosed globally each year, making it the most common cause of acute flaccid paralysis. GBS usually manifests as variable degrees of weakness in the limbs or muscles, reduced or absent deep tendon reflexes, and a distinct profile in the cerebrospinal fluid and electrodiagnostic testing. [1,2] This syndrome may present in a variety of forms that are considered GBS variants. Acute inflammatory demyelinating polyneuropathy (AIDP), acute motor axonal neuropathy (AMAN), and acute motor and sensory axonal neuropathy (AMSAN) are the most prevalent of these subgroups. In addition, several other subtypes have been reported [3].

Since infection is usually the underlying cause, the incidence varies depending on the spread of infectious diseases, which are typically upper respiratory infections and gastrointestinal infections. Campylobacter jejuni, Cytomegalovirus, Epstein-Barr virus, and Zika virus are the most often occurring infections before the onset of GBS. In addition, GBS may occur following the administration of vaccines, such as the swine-influenza vaccine and seasonal influenza vaccine [4]. Our best understanding of its pathogenesis comes from research on individuals with specific infections, which have demonstrated that the disease is caused by the production of autoimmune antibodies and inflammatory cells that cross-react with epitopes on peripheral nerves and roots, resulting in demyelination, axonal injury, or both [5–7]. However; it is suggested that demyelination can be brought about by macrophage-mediated phagocytosis of peripheral nerve components [8, 9].

Recently, in the era of the coronavirus disease of 2019 (COVID-19) pandemic, there was an increase in the number of case reports of GBS associated with severe acute respiratory syndrome coronavirus 2 (SARS-COV-2) infection during the pandemic [10]. Not only that, COVID-19 vaccines have been also associated with GBS. In this systematic review, we aimed to investigate the incidence of GBS after COVID-19 vaccination, and describe its clinical characteristics and potential confounders.

2. Methods

2.1. Search strategy and study selection

The present research was conducted in according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [11], and pre-registered protocol on PROSPERO [CRD42022373086] that registered at 07/11/2022. Since our study did not involve human subjects, the need for an institutional review board (IRB) approval was exempted. An electronic search was conducted through four databases: PubMed, Scopus, medRxiv, and Google Scholar for all case reports and case series describing GBS manifestation in relation to COVID-19 vaccine administration. All published articles from inception until November 1st, 2022 were included. The search including the following keywords “COVID-19 vaccination and GBS”, “SARS and GBS”, AND “MERS and GBS”. The search criteria were adjusted as per the searched. A manual search was also conducted to avoid missing any potentially relevant studies.

3. Eligibility Criteria

Included studies had to satisfy the following criteria: case studies (case reports, and case series) discussing GBS incidence after administering COVID-19 vaccination. Article language was restricted to English. Study designs other than case reports and case series were excluded. Duplicate studies, studies with missing clinical data, unavailable full-texts, and articles unrelated to our study objective were excluded. In addition, adverse neurological manifestation other than GBS were excluded.

4. Selection of studies

Retrieved articles from electronic databases were imported in Endnote Software for duplicate removal. Then, citations were exported an Excel Sheet for screening. Screening was done on two steps: title/abstract and full text screening. Two sets of two reviewers each [AJT, YAK, OF, ITA and EMH] carried out the screening process, where the titles, abstracts, and full texts of all records were screened simultaneously by four reviewers. Differences between reviewers were solved by a thorough discussion, and when necessary, the senior author [MTA] was consulted to give a final decision in unsolved disputes.

5. Data extraction

Four reviewers [MTA, MJJT, AJT, YAK, and EMH] developed the data extraction sheet with the use of Google Forms. From the selected studies, we pulled out the following variables for our analysis: study type, date of publication, country of case origin, age, gender, the clinical presentation of GBS and its variants (including paraparesis/quadruparesis, cranial nerve deficits, and diarrhea), type of vaccine, number of doses, time of doses, site of injection, latency between COVID-19 vaccination and initial symptoms of GBS, cerebrospinal fluid (CSF) total protein levels, anti-ganglioside antibodies testing, imaging findings, Electromyography (EMG) & Nerve Conduction Studies (NCS) findings, treatments including standard commercially available Intravenous immunoglobulin (IVIG), Plasmapheresis (PLEX), chloroquine, hydroxychloroquine (HCQ), azithromycin, Interleukin-6 (IL-6) blockers (tocilizumab), corticosteroids, any other medications (paracetamol, aspirin, antibiotics, Non-steroidal anti-inflammatory drugs (NSAIDs), or any herbal).

6. Quality Assessment

The critical appraisal checklist for case reports provided by the National Institutes of Health (NIH) Quality assessment tool was used to perform an assessment of the overall quality of case series and case reports [12].

7. Data Synthesis

Brighton electrophysiological criteria and mortality outcomes were used to assess the confidence in GBS diagnosis in patients [13]. Moreover, Modified Erasmus GBS Outcome Score (mEGOS) was calculated for cases with sufficient information using Medscape's online calculator [14]. Qualitative and quantitative analysis were performed on acquired data. Frequencies and proportions were used to summarize the data. Categorical variables were compared using the Pearson Chi-square test. Statistical significance was deemed to exist at a p-value of 0.05. Survival analysis for the risk of GBS complication into need for mechanical ventilation or walking impairment were conducted, with each vaccine type and vaccine dose separately handled. The Social Sciences Statistical Program was used to conduct the statistical analysis (IBM SPSS Corp, Statistical Product and Service Solutions (SPSS) Statistics version 26, Chicago, USA).

8. Results

8.1. Database search results

A PRISMA diagram, showing the details of the database search and screening processes is shown in Fig. 1. Following the retrieval of studies from the initial database search, eight duplicated records were found and removed through EndNote Software. Then, the titles and abstracts of 207 records were screened against our eligibility criteria, yielding 148 records that warranted further full text screening. Of note, manual search did not yield any more relevant studies. Forty-five records were

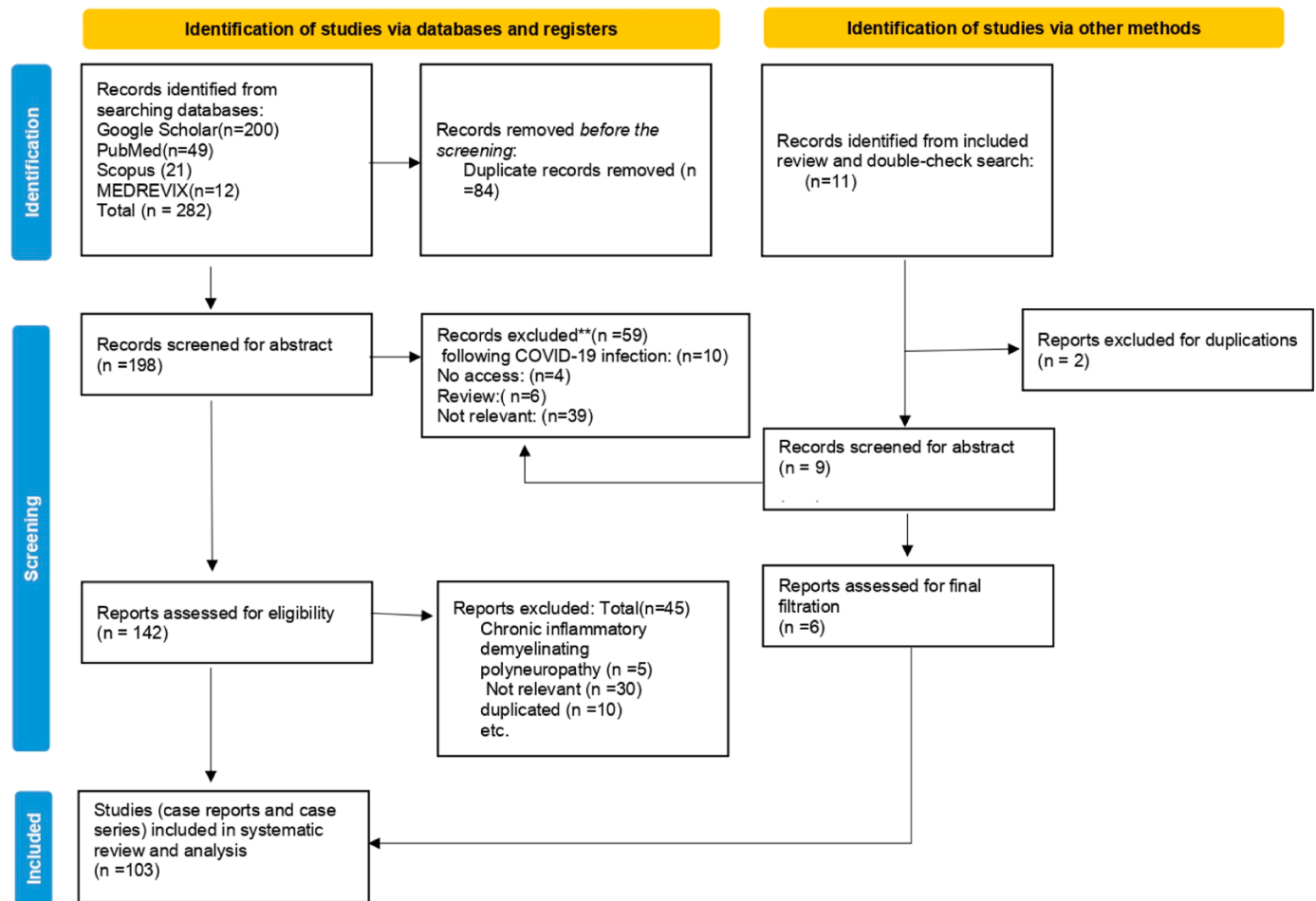


Fig. 1. PRISMA chart of article selection.

excluded, and the reasons for exclusion are provided in Fig. 1. Finally, a total of 103 studies [15–117] were included in the qualitative and quantitative synthesis of our review.

9. Baseline characteristics of included studies

A detailed description of the baseline characteristics of included studies is provided in Table 1. A total of 103 case reports and series with 175 cases of GBS following COVID-19 vaccination from 34 different countries were reported in this systematic review. Fig. 2 shows the distribution of the included studies over the world. According to the NIH quality assessment tool, the majority of the studies also had overall good quality.

The demographic details of the included cases are shown in Table 2 and subdivided by respective GBS subtype. The patients' ages ranged from 13 to 90, with an average age of (53.59 ± 18.83), and male cases (n=106, 60.57%) occurred 1.5 times more frequently than female cases (n=69, 39.43%). With 74 cases (42.29%), the AIDP subtype was the most frequently reported subtype. Age differences across the 5 categories were significantly different ($p = 0.049$). However, the AIDP and acute motor axonal neuropathy (AMAN) subtypes were primarily related to age group (54.84 ± 17.83 and 51.07 ± 20.77 , respectively), whereas the AMSAN subtype was primarily associated with older age (63.88 ± 20.87). In comparison to other vaccines, the AstraZeneca vaccine had the highest association with AIDP (n=38, 21.71%), followed by the Pfizer vaccine (n=23, 13.14%). It is important to note that the bilateral Facial Palsy (BFP) subtype was mostly linked to adenoviral vector vaccinations, since out of 22 cases of BFP, 14 were linked to

AstraZeneca and 2 to Johnson & Johnson, accounting for an average of 72% of the total BFP cases. In terms of dosage effect, the majority of GBS occurrences occurred exclusively after the administration of the first dose 135 (77.14%). Fortunately, the majority of GBS patients survived (n=168, 96%, $p=0.036$). However, despite the higher survival rate, most patients suffered from low mEGOS scores (4 ± 3.57). Table 3 shows the pattern of symptomatology among four GBS-subtype groups (AIDP, AMSAN, AMAN, and Others). Hypertension was the most common underlying comorbidity among cases, followed by diabetes mellitus (DM). However, the medication history was insignificant. Fig. 3

10. Unveiling the Clinical Presentation: Understanding the Symptoms of Patients

Most patients were characterized by symmetrical weakness (93.71%) with a greater proportion of AIDP-GBS variant patients reporting proximal (26.86%, $p<0.01$) and distal limb weakness (31.43%, $p=0.667$) compared to other subtypes. In addition to weakness in limbs, facial paralysis (42.3%) and areflexia (50.1%) were also common motor findings. Regarding the associated sensory complications, dyesthesias were the most reported (60%), especially among the AIDP subtype. Modified Medical Research Council (MRC) muscle power score was calculated, and the MRC score was highest in the AIDP patients (41.36 ± 13.75) while the lowest score was associated with AMSAN-GBS (30.55 ± 12.66). Albuminocytological dissociation was present in 127 out of 175 GBS cases (72.57%). Among these, AIDP was the most common variant. In addition, we assessed whether the patients fulfilled the Brighton criteria for diagnosis of GBS. We found that most of our cohort

Table 1

Baseline characteristics of included studies (N = 103).

No.	Author	Year	Country	Type	No. of Patients	Age	Sex	GBS Subtype	Quality
1	Abicic et al. [13]	2021	Croatia	Case Report	1	24	F	MFS	Good
2	Abo-Zed & Pinevich [14]	2021	N/A	Case Report	1	87	M	N/A	Poor
3	Aldeeb et al. [15]	2022	Qatar	Case Report	1	81	F	AMSAN	Good
4	Alekseyev et al. [16]	2022	USA	Case Report	1	34	F	N/A	Good
5	Aljamea et al. [17]	2022	Saudi Arabia	Case Series	2	46	2 M	AMAN, BBE	Good
6	AlKolfat et al. [18]	2022	Egypt	Case Report	1	23	M	N/A	Fair
7	Allen et al. [19]	2021	UK	Case Series	4	46.5	4 M	4 BFP	Fair
8	Aly et al. [20]	2022	Egypt	case Report	1	29	M	AIDP	Good
9	Andreozzi et al. [21]	2022	Italy	Case Series	2	51	F, M	BFP, AIDP	Good
10	Anjum et al. [22]	2022	USA	Case Report	1	80	M	AIDP	Good
11	Arce Gálvez et al. [23]	2021	Colombia	Letter to Editor	1	73	M	AIDP	Good
12	Azam et al. [24]	2021	UK	Case Report	1	67	M	AIDP	Good
13	Bax et al. [25]	2021	Italy	Letter to Editor	2	70.5	F, M	AMSAN, AIDP	Good
14	Bazrafshan et al. [26]	2022	Iran	Case Report	1	68	F	AIDP	Poor
15	Bello Marq & Aguirre III [27]	2022	Philippines	Case Report	1	51	M	AMAN	Good
16	Bellucci et al. [28]	2022	Italy	Case Report	1	57	M	AIDP	Good
17	Benlamkadam et al. [29]	2022	Morocco	Case Series	2	69	2 F	AMSAN, AIDP	Poor
18	Biswas et al. [30]	2021	India	Case Report	1	49	M	N/A	Poor
19	Bonifacio et al. [31]	2021	UK	Case Series	5	56.8	4 M, F	5 BFP	Good
20	Bouattour et al. [32]	2021	Tunisia	Case Report	1	67	M	AIDP	Good
21	Carranza et al. [33]	2022	USA	Case Report	1	53	F	AIDP	Good
22	Castiglione et al. [34]	2022	Argentina	Case Series	9	54.33	7 M, 2 F	5 BFP, 4 AIDP	Good
23	Censcak et al. [35]	2021	Czech	Case Report	1	42	M	AIDP	Good
24	Chang [36]	2022	South Korea	Case Report	1	32	M	N/A	Good
25	Chen et al. [37]	2022	USA	Case Report	1	75	F	MFS	Fair
26	Chun et al. [38]	2021	South Korea	Case Series	2	78	2 F	AMSAN, AIDP	Good
27	Da Silva et al. [39]	2021	Brazil	Case Report	1	62	F	N/A	Good
28	Dalwadi et al. [40]	2021	USA	Case Report	1	86	F	AMAN	Good
29	Dang & Bryson [41]	2021	Australia	Case Report	1	64	M	MFS	Poor
30	Demissie et al. [42]	2022	Ethiopia	Case Report	1	45	M	AMAN	Good
31	Donaldson & Margolin [43]	2022	Canada	Case Report	1	45	M	AIDP	Poor
32	Fakhari et al. [44]	2022	Iran	Case Report	1	60	M	AMSAN	Good
33	Fernandes et al. [45]	2022	Trinidad and Tobago	Case Series	1	50	M	AIDP	Good
34	Finsterer [47]	2021	Austria	Case Report	1	32	M	AIDP	Good
35	Finsterer [46]	2021	Austria	Case Report	1	69	F	AIDP	Good
36	Fukushima et al. [48]	2022	Japan	Case Report	1	55	F	Sensory GBS	Poor
37	Germano et al. [49]	2022	Italy	Case Series	13	64.07	9 M, 4 F	11 AIDP, 2 AMSAN	Good
38	Gonzales et al. [50]	2022	Philippines	Case Report	1	68	M	AIDP	Fair
39	Gunawan et al. [51]	2022	Indonesia	Case Report	1	14	M	AIDP	Fair
40	Hasan et al. [52]	2021	UK	Case Report	1	62	F	AIDP	Good
41	Hemalatha et al. [53]	2021	India	Case Report	1	43	F	AMAN	Good
42	Hilts et al. [54]	2022	USA	Case Report	1	58	M	N/A	Good
43	Hiraga et al. [55]	2022	Japan	Case Report	1	19	M	BFP	Good
44	Hughes et al. [56]	2021	USA	Case Report	1	65	M	AIDP	Good
45	Ilyas et al. [57]	2022	USA	Case Report	1	62	M	N/A	Good
46	Introna et al. [58]	2021	Italy	Case Report	1	62	M	N/A	Fair
47	Jain et al. [59]	2021	USA	Case Report	1	65	F	BFP	Good
48	Jain et al. [60]	2022	India	Letter to Editor	2	48.5	F, M	2 AMSAN	Fair
49	Kanabar et al. [61]	2021	UK	Case Series	2	58.5	F, M	2 AIDP	Good
50	Karimi et al. [62]	2021	Iran	Case Series	9	54.22	7 M, 2 F	3 ADIP, 2 AMAN, 2 AMSAN, 2 N/A	Good
51	Khadka et al. [63]	2021	Nepal	Case Report	1	44	M	AIDP	Fair
52	Khanal et al. [64]	2021	Nepal	Case Report	1	50	F	N/A	Fair
53	Kim et al. [67]	2021	South Korea	Case Report	1	21	M	BFP	Good
54	Kim et al. [68]	2022	USA	Case Report	1	16	F	Sensory GBS	Good
55	Kim et al. [66]	2021	South Korea	Case Series	2	45	F, M	2 AIDP	Good
56	Kim et al. [65]	2022	South Korea	Case Series	13	56.38	9 F, 4 M	3 AMAN, 3 Paraparetic GBS, 3 AIDP, Acute Cervicobrachial GBS, 3 N/A	Good
57	Koreen et al. [69]	2021	USA	Case Report	1	59	F	N/A	Good
58	Kripalani et al. [70]	2021	India	Case Report	1	52	F	AIDP	Poor
59	Kulsirichawaroj et al. [71]	2022	Thailand	Case Report	1	16	F	Polyneuritis Cranialis	Fair
60	Lanman et al. [72]	2022	USA	Case Report	1	58	F	AMSAN	Good
61	Lee et al. [73]	2021	Korea	Case Report	1	61	M	AMSAN	Good

(continued on next page)

Table 1 (continued)

No.	Author	Year	Country	Type	No. of Patients	Age	Sex	GBS Subtype	Quality
62	Leemans et al. [74]	2022	Belgium	Case Series	4	66.75	2 F, 2 M	3 AIDP, BFP	Good
63	Liang et al. [75]	2022	China	Case Report	1	64	M	MFS	Fair
64	Ling et al. [76]	2021	Canada	Case Report	1	63	M	N/A	Good
65	Luitel et al. [77]	2022	Nepal	Case Report	1	48	M	AIDP	Good
66	Malamud et al. [78]	2022	USA	Case Report	1	14	M	AIDP	Good
67	Maramattom et al. [79]	2021	India	Case Series	7	62.71	6 F, M	5 AIDP, 2 N/A	Good
68	Masuccio et al. [80]	2021	Italy	Letter to Editor	1	N/A	M	AMSAN	Fair
69	Matarneh et al. [81]	2021	Qatar	Case Report	1	61	M	AIDP	Good
70	McKean & Chircop [82]	2021	Malta	Case Report	1	48	M	AIDP	Good
71	Min et al. [83]	2021	South Korea	Case Series	2	47.5	F, M	2 Sensory GBS	Good
72	Morehouse et al. [84]	2021	USA	Case Report	1	49	F	AMSAN	Good
73	Mosna et al. [85]	2022	Slovakia	Case Report	1	71	M	AIDP	Good
74	Mozdhehipanah et al. [86]	2021	Iran	Case Report	1	32	M	AIDP	Good
75	Nagalli & Kikkeri [87]	2022	USA	Case Report	1	49	F	N/A	Good
76	Nagdev et al. [88]	2022	India	Case Series	5	34	4 M, F	3 AIDP, 2 AMSAN	Poor
77	Nanatsue et al. [89]	2022	Japan	Case Report	1	72	M	MFS	Fair
78	Nasiri et al. [90]	2022	Saudi Arabia	Case Series	2	30	2 M	2 N/A	Fair
79	Nasuelli et al. [91]	2021	Italy	Case Report	1	59	M	AIDP	Good
80	Nishiguchi et al. [92]	2021	Japan	Case Report	1	71	M	MFS	Good
81	Ogata et al. [93]	2022	Japan	Case Report	1	70	M	AIDP	Good
82	Ogbebor et al. [94]	2021	USA	Case Report	1	86	F	N/A	Good
83	Patel et al. [95]	2021	UK	Case Report	1	37	M	N/A	Good
84	Pirola et al. [96]	2022	Brazil	Case Report	1	47	F	MFS	Fair
85	Prado et al. [97]	2022	Philippines	Case Report	1	35	M	AIDP	Poor
86	Prasad et al. [98]	2021	USA	Case Report	1	41	M	N/A	Good
87	Rao et al. [99]	2021	USA	Case Report	1	42	F	AIDP	Good
88	Razok et al. [100]	2021	Qatar	Case Report	1	73	M	AIDP	Good
89	Richardson-May et al. [101]	2022	UK	Case Report	1	71	F	BFP	Fair
90	Rohilla et al. [102]	2022	India	Case Report	1	13	F	AMAN	Fair
91	Rossetti et al. [103]	2021	USA	Case Report	1	38	M	BFP	Good
92	Saraswat et al. [104]	2021	India	Case Report	1	29	M	N/A	Fair
93	Scendoni et al. [105]	2021	Italy	Case Report	1	82	F	AIDP	Good
94	Siddiqi et al. [106]	2022	Pakistan	Case Report	1	53	M	MFS	Fair
95	Sosa-Hernández et al. [107]	2022	Mexico	Case Report	1	23	M	AIDP	Good
96	Su et al. [108]	2022	Taiwan	Case Report	1	39	F	AIDP	Good
97	Tabatabaee et al. [109]	2022	Iran	Case Series	3	38	3 M	3 AMAN	Good
98	Takahashi et al. [110]	2022	Japan	Case Report	1	65	F	AIDP	Good
99	Thant et al. [111]	2022	USA	Case Report	1	66	M	N/A	Good
100	Trimboli et al. [112]	2021	Italy	Case Report	1	25	F	N/A	Good
101	Waheed et al. [113]	2021	USA	Case Report	1	82	F	N/A	Fair
102	Yamakawa et al. [114]	2021	Japan	Case Report	1	30	M	MFS	Fair
103	Zubair et al. [115]	2022	USA	Case Report	1	30	F	BFP	Good

AIDP: Acute inflammatory demyelinating polyradiculoneuropathy, AMAN: Acute motor axonal neuropathy, AMSAN: Acute motor-sensory axonal neuropathy, BBE: Bickerstaff's brainstem encephalitis, BFP: Bilateral weakness with paresthesia, MFS: Miller Fisher syndrome.

satisfied the level 2 Brighton criteria (54.29%). However, Level 4 was the least reported among cases with only 4 cases. Among these presentations, AIDP was the one with the most manifestations in all these categories. Whereas, the other 2 main variants resulted in a similar frequency of manifestations in most categories, except in dysesthesias where the AMAN was six-fold more frequent than AMSAN. (Table 3).

On average, patients began to have symptoms of GBS 13.43 ± 11.45 days from vaccination ($p=0.73$), and survival analysis for complication of GBS into mechanical ventilation or walking impairment yielded a significant drop in survival around 13 days, and a severely increased probability of complication after 25 days, especially for Pfizer-BioNTech, Moderna, and AstraZeneca vaccines ($p<0.01$). Kaplan-Meier survival analysis curves classified according to vaccine type and vaccine dose are shown in Figs. 4 and 5, respectively.

11. Management Approach

Table 4 shows the case management approaches that were used. We divided them into five different categories of interventions. The most often used intervention and the only one with a statistically significant

link across GBS-subtypes ($p=0.03$) was intravenous immune globulin (IVIg). Rehabilitation was applied in 38 patients (21.7% of the cases) as an adjuvant therapy.

12. Discussion

In this systematic review, cases of GBS after COVID-19 vaccination have been collected and analyzed in an attempt to describe the prevalence and characteristics of GBS following COVID-19 vaccines. Over 69% of the world population received at least one dose of COVID-19 vaccine. [118] Our cohort included patients from several countries worldwide. A clinical correlation could be established between geographical location and GBS through the Human leukocyte antigens (HLA-DQB1) and anti-glycosphingolipid antibodies, as they have been linked to the severity of GBS in literature [119]. However, a study with a large enough population is yet to be done, and available studies are mostly local or small sample size.

A higher percentage of males was affected (60.57%). GBS is epidemiologically known to affect males more than females [120,121]. Males, despite the argument that they are often underdiagnosed in cases of

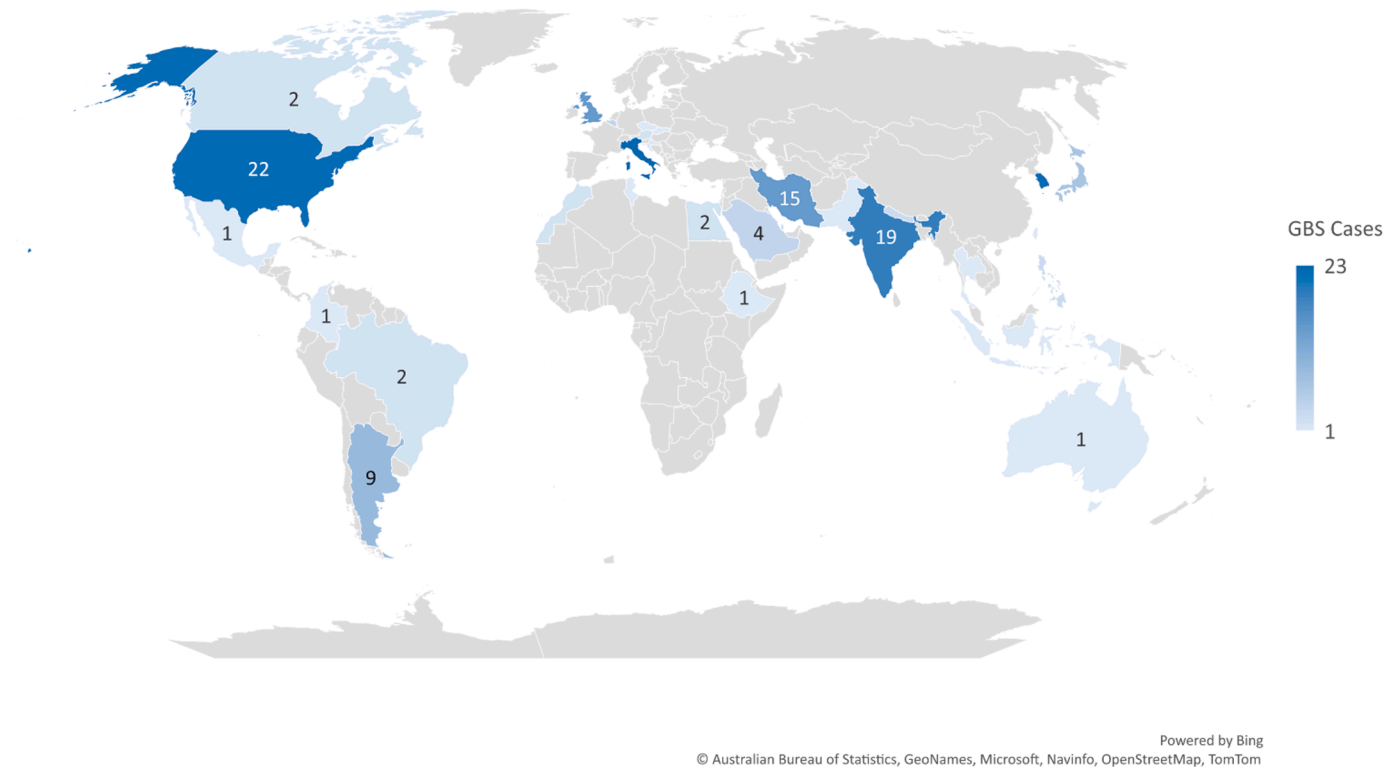


Fig. 2. Distribution of acquired studies over the world.

Table 2
Demographic characteristics of the included cases.

Characteristic		GBS-Subtype					Total	p-value
		AIDP	AMAN	AMSAN	Other	N/A		
Demographics								
No. of Patients		74 (42.29%)	14 (8%)	17 (9.71%)	42 (24%)	28 (16%)	175 (100%)	
Age (mean ± SD)		54.84 ±17.83	51.07 ±20.77	63.88 ±20.87	49.6 ±18.55	51.64 ±18.43	53.59 ±18.83	0.049
Gender								0.889
	Female	28 (16%)	4 (2.29%)	7 (4%)	18 (10.29%)	12 (6.86%)	69 (39.43%)	
	Male	46 (26.29%)	10 (5.71%)	10 (5.71%)	24 (13.71%)	16 (9.14%)	106 (60.57%)	
Vaccine Type								0.02
	Oxford/AstraZeneca	38 (21.71%)	6 (3.43%)	7 (4%)	20 (11.43%)	12 (6.86%)	83 (47.43%)	
	Pfizer-BioNTech	23 (13.14%)	1 (0.57%)	5 (2.86%)	14 (8%)	7 (4%)	50 (28.57%)	
	Johnson & Johnson	2 (1.14%)	1 (0.57%)	1 (0.57%)	2 (1.14%)	4 (2.29%)	10 (5.71%)	
	Moderna	2 (1.14%)	1 (0.57%)	2 (1.14%)	1 (0.57%)	3 (1.71%)	9 (5.14%)	
	Corbevax	0 (0%)	1 (0.57%)	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	
	Covaxin	1 (0.57%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	
	Sinopharm BIBP	1 (0.57%)	3 (1.71%)	2 (1.14%)	0 (0%)	0 (0%)	6 (3.43%)	
	Sinovac-Coronavac	3 (1.71%)	1 (0.57%)	0 (0%)	2 (1.14%)	0 (0%)	6 (3.43%)	
	Sputnik V	4 (2.29%)	0 (0%)	0 (0%)	3 (1.71%)	2 (1.14%)	9 (5.14%)	
Vaccine Dose								0.576
	First	54 (30.86%)	12 (6.86%)	13 (7.43%)	33 (18.86%)	23 (13.14%)	135 (77.14%)	
	Second	20 (11.43%)	2 (1.14%)	3 (1.71%)	8 (4.57%)	4 (2.29%)	37 (21.14%)	
	Booster	0 (0%)	0 (0%)	1 (0.57%)	1 (0.57%)	1 (0.57%)	3 (1.71%)	
Mechanical Ventilation								0.339
	Ventilated	11 (6.29%)	2 (1.14%)	2 (1.14%)	1 (0.57%)	3 (1.71%)	19 (10.86%)	
	Not Ventilated	63 (36%)	12 (6.86%)	15 (8.57%)	41 (23.43%)	25 (14.29%)	156 (89.14%)	
Outcome								0.036
	Survived	71 (40.57%)	14 (8%)	14 (8%)	41 (23.43%)	28 (16%)	168 (96%)	
	Died	3 (1.71%)	0 (0%)	3 (1.71%)	1 (0.57%)	0 (0%)	7 (4%)	
mEGOS score (mean ± SD)		4.32 ±3.41	5.79 ±3.72	5.53 ±3.89	1.6 ±1.98	4.93 ±3.95	4 ±3.57	0.003

AIDP: Acute inflammatory demyelinating polyradiculoneuropathy, AMAN: Acute motor axonal neuropathy, AMSAN: Acute motor-sensory axonal neuropathy, mEGOS score: modified Erasmus Guillain-Barré Syndrome Outcome Score.

autoimmunity, are thought to have different immune mechanisms than females, rendering them more susceptible to infections, likely due to an inferior regulatory T cells response [122–124]. In addition, difference in immune-regulation between sexes have long been reported in literature,

and in the context of GBS, genes involved in matrix metalloproteinase-9 (MMP9) production, which is associated with increased GBS severity, were expressed more in males [125]. Another potential correlation between GBS and males has to do with the “target organ resistance”.

Table 3
Patterns of symptomatology among affected cases.

Characteristic	GBS Subtype					Total	p-Value
	AIDP	AMAN	AMSAN	Other	N/A		
COVID-19 PCR							
Negative	44 (25.14%)	10 (5.71%)	14 (8%)	26 (14.86%)	22 (12.57%)	116 (66.29%)	0.387
Positive	1 (0.57%)	0 (0%)	0 (0%)	2 (1.14%)	1 (0.57%)	4 (2.29%)	
N/A	29 (16.57%)	4 (2.29%)	3 (1.71%)	14 (8%)	5 (2.86%)	55 (31.43%)	
Underlying Comorbidities							
Hypertension	13 (7.43%)	3 (1.71%)	5 (2.86%)	6 (3.43%)	11 (6.29%)	38 (21.71%)	0.1
DM	5 (2.86%)	1 (0.57%)	3 (1.71%)	3 (1.71%)	4 (2.29%)	16 (9.14%)	
Previous GBS	3 (1.71%)	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	4 (2.29%)	0.566
Other	15 (8.57%)	3 (1.71%)	8 (4.57%)	8 (4.57%)	8 (4.56%)	42 (24%)	
Medications History							
Antihypertensives	3 (1.71%)	0 (0%)	3 (1.71%)	2 (1.14%)	2 (1.14%)	10 (5.71%)	0.207
Antiviral Drugs	1 (0.57%)	0 (0%)	0 (0%)	1 (0.57%)	0 (0%)	2 (1.14%)	
NSAIDs	1 (0.57%)	0 (0%)	1 (0.57%)	2 (1.14%)	1 (0.57%)	5 (2.86%)	0.703
Diuretics	1 (0.57%)	0 (0%)	2 (1.14%)	0 (0%)	0 (0%)	3 (1.71%)	
Chemotherapy	1 (0.57%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	0.849
DM Drugs	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	1 (0.57%)	
Other	3 (1.71%)	0 (0%)	1 (0.57%)	4 (2.29%)	0 (0%)	8 (4.57%)	0.344
Time between COVID-vaccine and GBS manifestation (days)	12.03 ±7.74	10.29 ±9.14	13.18 ±11.09	14.88 ±13.64	16.68 ±16.22	13.43 ±11.45	
Diarrhea before onset?							0.614
N/A	44 (25.14%)	6 (3.43%)	9 (5.14%)	27 (15.43%)	12 (6.86%)	98 (56%)	0.63
No	28 (16%)	7 (4%)	8 (4.57%)	14 (8%)	14 (8%)	71 (40.57%)	
Yes	2 (1.14%)	1 (0.57%)	0 (0%)	1 (0.57%)	2 (1.14%)	6 (3.43%)	
Symmetry of disease							<0.01
Asymmetrical	6 (3.43%)	0 (0%)	0 (0%)	3 (1.71%)	2 (1.14%)	11 (6.29%)	
Symmetrical	68 (38.86%)	14 (8%)	17 (9.71%)	39 (22.29%)	26 (14.86%)	164 (93.71%)	
Symptomatology							
Motor Symptoms							
Proximal limb Weakness	47 (26.86%)	10 (5.71%)	15 (8.57%)	15 (8.57%)	15 (8.57%)	102 (58.29%)	<0.01
Distal limb Weakness	55 (31.43%)	13 (7.43%)	13 (7.43%)	15 (8.57%)	25 (14.29%)	121 (69.14%)	
Neck Flexion Weakness	1 (0.57%)	0 (0%)	0 (0%)	2 (1.14%)	1 (0.57%)	4 (2.29%)	0.1
Hyporeflexia	20 (11.43%)	5 (2.86%)	1 (0.57%)	6 (3.43%)	4 (2.29%)	36 (20.57%)	
Areflexia	40 (22.86%)	7 (4%)	8 (4.57%)	12 (6.86%)	22 (12.57%)	89 (50.86%)	<0.01
Ocular Motility Impairment	5 (2.86%)	0 (0%)	1 (0.57%)	14 (8%)	1 (0.57%)	21 (12%)	
Facial Paralysis (Partial/Complete)	31 (17.71%)	2 (1.14%)	3 (1.71%)	29 (16.57%)	9 (5.14%)	74 (42.29%)	<0.01
Sensory Symptoms							
Dysesthesias	53 (30.29%)	2 (1.14%)	12 (6.86%)	23 (13.14%)	15 (8.57%)	105 (60%)	0.349
Pain	13 (7.43%)	2 (1.14%)	4 (2.29%)	14 (8%)	7 (4%)	40 (22.86%)	
Dysgeusia	1 (0.57%)	0 (0%)	0 (0%)	1 (0.57%)	0 (0%)	2 (1.14%)	0.061
N/A	18 (10.29%)	10 (5.71%)	5 (2.86%)	14 (8%)	10 (5.71%)	57 (32.57%)	
Autonomic Symptoms							
Dysautonomia	8 (4.57%)	3 (1.71%)	6 (3.43%)	5 (2.86%)	4 (2.29%)	26 (14.86%)	0.876
Dyspnea	7 (4%)	1 (0.57%)	1 (0.57%)	2 (1.14%)	3 (1.71%)	14 (8%)	
Dysphagia	0 (0%)	0 (0%)	0 (0%)	1 (0.57%)	2 (1.14%)	3 (1.71%)	0.667
SIADH	1 (0.57%)	1 (0.57%)	0 (0%)	1 (0.57%)	1 (0.57%)	4 (2.29%)	
Dysphagia	4 (2.29%)	2 (1.14%)	3 (1.71%)	1 (0.57%)	5 (2.86%)	15 (8.57%)	0.441
Bowel Disorder	3 (1.71%)	1 (0.57%)	2 (1.14%)	1 (0.57%)	3 (1.71%)	10 (5.71%)	
N/A	50 (28.57%)	9 (5.14%)	9 (5.14%)	34 (19.43%)	15 (8.57%)	117 (66.86%)	0.171
Anti-bodies							
anti-GM1	3 (1.71%)	1 (0.57%)	0 (0%)	3 (1.71%)	1 (0.57%)	8 (4.57%)	0.384
anti-GD1A	3 (1.71%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	3 (1.71%)	
anti-GT1A	1 (0.57%)	0 (0%)	1 (0.57%)	2 (1.14%)	0 (0%)	4 (2.29%)	<0.01
anti-GQ1B	1 (0.57%)	1 (0.57%)	1 (0.57%)	7 (4%)	0 (0%)	10 (5.71%)	
CSF Analysis							
Protein concentration (g/L)	2.52 ±4.24	1.08 ±0.68	1.54 ±1.03	1.89 ±2.41	11.47 ±31.21	3.49 ±12.6	0.659
Albumino-cytological dissociation							
Absent	8 (4.57%)	2 (1.14%)	0 (0%)	3 (1.71%)	2 (1.14%)	15 (8.57%)	<0.01
Present	56 (32%)	9 (5.14%)	13 (7.43%)	31 (17.71%)	18 (10.29%)	127 (72.57%)	
MRC score	41.36 ±13.75	35.38 ±16.19	30.55 ±12.66	51.2 ±12	36.79 ±17.2	40.06 ±15.22	0.77
Brighton's Criteria							
1	28 (16%)	2 (1.14%)	3 (1.71%)	2 (1.14%)	6 (3.43%)	41 (23.43%)	0.012
2	35 (20%)	8 (4.57%)	11 (6.29%)	28 (16%)	13 (7.43%)	95 (54.29%)	
3	11 (6.29%)	4 (2.29%)	3 (1.71%)	9 (5.14%)	8 (4.57%)	35 (20%)	
4	0 (0%)	0 (0%)	0 (0%)	3 (1.71%)	1 (0.57%)	4 (2.29%)	
Total	74 (42.29%)	14 (8%)	17 (9.71%)	42 (24%)	28 (16%)	175 (100%)	

AIDP: Acute inflammatory demyelinating polyradiculoneuropathy, AMAN: Acute motor axonal neuropathy, AMSAN: Acute motor-sensory axonal neuropathy, DM: Diabetes Mellitus, PCR: Polymerase chain reaction, SIADH: Syndrome of inappropriate antidiuretic hormone secretion.

Peripheral nerve biology elicits differences between sexes, and myelin with higher vulnerability to inflammation might be behind this unusual pattern of autoimmunity [126]. Patients in our cases were on average in their late adulthood, with a mean age of (53.59 ±18.83) years.

Moreover, in literature, this age group has had the highest rates of COVID-19 vaccination complications [127]. Late adulthood has been showing the highest rate of COVID-19 vaccination, which may explain the increased complication rates in this age group [128]. Hanson KE

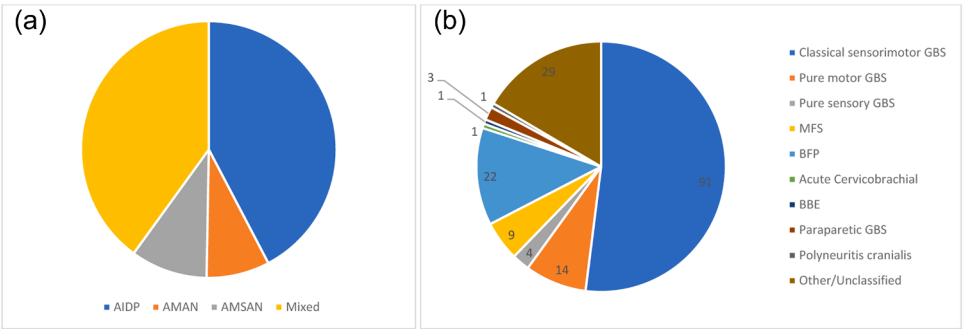


Fig. 3. Distribution of GBS subtypes across cases Fig. 3.A: Electrophysiologic-subtypes Fig. 3.B: Clinical-subtypes.

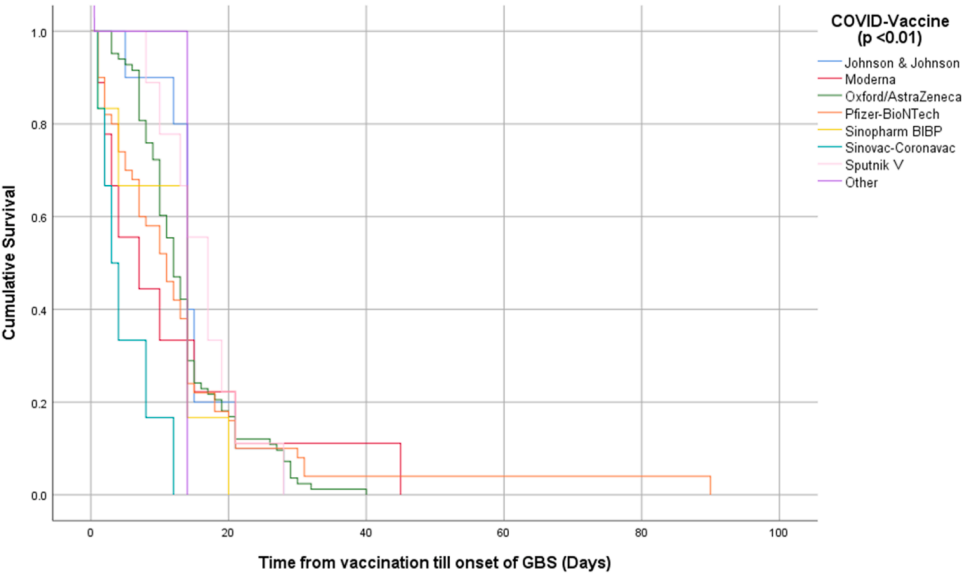


Fig. 4. Kaplan-Meier curve for survival analysis of each COVID-19 vaccine type.

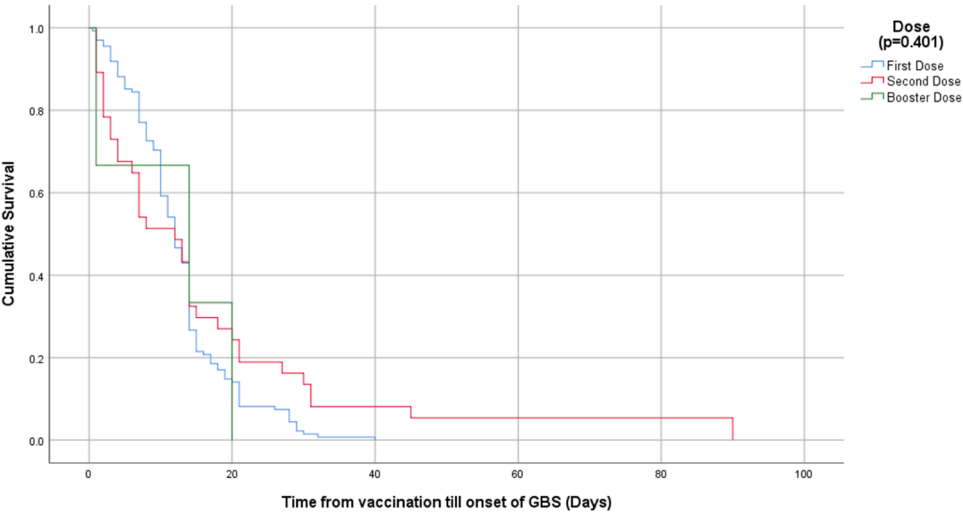


Fig. 5. Kaplan-Meier curve for survival analysis of each COVID-19 vaccine dosage.

et al., in a study conducted on GBS incidence post COVID-vaccines analyzed data on 7.8 million patients found similar results regarding both age and sex of affected individuals [129]. In their sample, average ages of GBS patients were 50 years for AstraZeneca recipients and 53 for Pfizer/Moderna recipients. Moreover, research have indicated that

males are probably more prevalent than females.

GBS electrophysiological subtypes vary in prevalence relative to geographic location, with AIDP dominating north America and Europe, and AMAN presenting more in Asia [130]. This difference in prevalence is often attributed to multivariate mechanism including genetic,

Table 4
Medical and surgical interventions applied for management of included cases.

Management	GBS Subtype					Total	p-Value
	AIDP	AMAN	AMSAN	Other	N/A		
IVIG	64 (36.6%)	9 (5.1%)	16 (9.1%)	28 (16%)	21 (12%)	138 (78.9%)	0.03
Plasmapheresis	5 (2.9%)	1 (0.6%)	1 (0.6%)	2 (1.1%)	5 (2.9%)	14 (8%)	0.335
Corticosteroids	8 (4.6%)	1 (0.6%)	2 (1.1%)	7 (4%)	3 (1.7%)	21 (12%)	0.858
Physical therapy/ Rehabilitation	17 (9.7%)	3 (1.7%)	5 (2.9%)	4 (2.3%)	9 (5.1%)	38 (21.7%)	0.19
Observation (No treatment)	4 (2.3%)	1 (0.6%)	0 (0%)	7 (4%)	2 (1.1%)	14 (8%)	0.166
Other	2 (1.1%)	0 (0%)	2 (1.1%)	8 (4.6%)	4 (2.3%)	16 (9.1%)	0.026
Total	74 (42.3%)	14 (8%)	17 (9.7%)	42 (24%)	28 (16%)	175 (100%)	

AIDP: Acute inflammatory demyelinating polyneuropathy; AMAN: Acute motor axonal neuropathy; AMSAN: Acute motor and sensory axonal neuropathy; IVIG: Intravenous immune globulin; N/A: Not assessed.

infectious and environmental factors [130]. In our cohort, the majority of cases were reported in North America and Europe, and correspondingly, AIDP (n=47, 42.29%) was reported most often, followed by AMAN and AMSAN (n=31, 17.71%). To add, axonal subtypes of GBS were observed more in older patients, especially AMSAN (63.88 ±20.87). This cohort is often diagnosed later than other age groups, likely due to existing comorbidities, that may mask the effects of GBS until the disease had progressed to motor-sensory involvement [131, 132].

BFP variant was reported more than expected. It is often regarded as a very rare subtype of GBS, but 22 cases (12.57%) of our cohort were diagnosed with this variant which indicate that COVID-19 vaccination is associated with more BFP-subtype incidence which could explained by the using the adenoviral vector mechanism [133]. This increase in the incidence of BFP-GBS with COVID-19 vaccination has been reported in literature, especially with the AstraZeneca vaccine [129,134,135], which could be attributed to the induced higher levels of specific T cells by adenovector vaccines [136]. This correlation between GBS and COVID-19 vaccination is not yet fully understood, however; pathophysiological mechanisms have been suggested. Molecular mimicry is suggested as the main explanation for this incidence, since the immunization induced by COVID-19 vaccines targets the virus’s spike proteins, which could bind sialic acid contained in gangliosides and glycoprotein on cell surface [60]. Another suggested mechanism is a complement deposition damaging the surface of Schwann cells [137]. In contrast, antibodies against self-antigens presents of Schwann cells are observed more with AIDP subtypes, which could explain the increased rated of AIDP incidence with AstraZeneca vaccines observed in our cohort (n=38, 21.71%), which relies on antigen-antibody complexes in exerting its effect [60,130]. Although anti-ganglioside autoantibodies have been suggested as a mechanism by which GBS manifests, sufficient data to support this claim is lacking, especially since many cases pass without antibodies testing [138]. To add, other antibodies that can cause demyelination or inflammatory neuropathies such as the neurofascin-155 and neurofascin-186 autoantibodies were not sufficiently explored in GBS related to COVID-19 or its vaccines [139]. However, using IVIG was the most commonly used intervention among our cohort.

Since all suggested mechanisms are immune-mediated, it was convenient that most cases of GBS manifest after the first dose of vaccines, as it triggers the immune reaction sustained by later doses [140]. GBS was reported with higher frequency with mRNA and adenovirus vector vaccines, and in the present study, 152 (86.8%) of patients received these vaccines. However, these findings need conformation, especially since data on each type of vaccine and recipients count are not available for comparison. Nevertheless, adenovirus vector vaccines, namely AstraZeneca and Johnson & Johnson, were associated with the highest incidence of GBS, and GBS was added to risks of adenoviral vaccines by the World Health Organization (WHO) and the European medicine agency [121,141,142].

In the present study, hypertension was the most prevalent

comorbidity to GBS (n=38, 21.71%), which is to be expected, since hypertension is one of the most common disorders worldwide, and no significant correlation is suspected between GBS and hypertension in our cohort [143].

MRC scores for cases included in this review were acquired. A higher score was recorded in AIDP patients (41.36 ±13.75) while the lower scores were associated with AMSAN-GBS (30.55 ±12.66). These scores are of important diagnostic and prognostic value, and while AIDP-GBS can progress to AMAN or AMSAN, either of which indicates a severe form of GBS, the MRC scores are expected to reflect the severity of the disease and its extent in patients [144]. González-Suárez et al. have demonstrated that axonal lesions corresponded with higher GBS severity [145].

Despite the low mortality rate in our cases, most patients had potential for GBS complication, and the average mEGOS score was on the lower end (4 ±3.57). Correspondingly, an average of level 2 was found using the Brighton criteria, with most patients assessed at levels 2 and 3 rather than 1 and 4. Early diagnosis of GBS can be of great importance, especially in reducing the risk of complication and applying needed management, preventing further deterioration [146]. COVID-19 lockdown caused a delayed presentation to clinics for several groups of diseased individuals [147–149]. In this view, we suspect that many GBS patients delayed their complaint due to COVID-19 related restrictions, especially in developing countries. Additionally, the temporal link between COVID-19 vaccination and the manifestation of GBS is of important prognostic value. On average, cases reported an onset of GBS around 13.43 ±11.45 days after vaccination. We performed survival analysis to assess the risk of GBS complication with walking impairment or the need for mechanical ventilation, and the risk increases drastically if the patient developed GBS more than 20 days from vaccination. Pfizer scientist had previously declared that the do not know what happens to immunity after 21 days of receiving their vaccine [150], and the exact mechanism of pathology afterwards is still in need of investigation. Moreover, a very interesting finding was detected between these numbers. It has been reported that the risk of needing mechanical ventilation in GBS patients increase in the first 7 days from onset to symptoms [7,130]. In our cohort, this is the same period at which the risk of needing mechanical ventilation increases dramatically. The 7 days between onset of GBS (13.43 days from vaccination) and the sharp drop in Kaplan-Meier survival curves.

Later GBS-diagnosis after COVID-19 vaccination was linked with higher clinical severity in literature [140], which further emphasizes the importance of urgent and quick diagnosis and management of GBS cases. When comparing our findings to studies that examined at GBS after COVID-19 infection, we found that age, gender, latency between the event and disease, the GBS EMG variation, the management strategy, the Brighton criteria, and albumino-cytological dissociation were similar. Contrarily, after COVID-19 infection, BPF occurrences decreased and antiganglioside antibody positive cases were considerably more prevalent (25 cases in our results compared to only 3 positive cases after COVID-19 infection) [10].

13. Recommendations and limitations

This study tried to investigate the association between GBS and COVID-19 vaccination. However, several limitations remain. Unfortunately, detailed information regarding vaccination demographics, which could have improved the results of this study, are not available. Moreover, several cases of GBS are not sufficiently documents, and diagnosis are often made without the necessary investigation and testing. To add, data on follow up on patients is scarce, which impairs the ability to adequately assess the course and outcome of the disease.

Although recorded, cases of GBS after COVID-19 vaccination remain relatively few, and vaccination against COVID-19 is not to be postponed on such a note. It is important, however, that physicians remain aware about the possibility of developing GBS after vaccine administration. Moreover, vaccine recipients should not hesitate to report any unusual findings after COVID-19 vaccination.

14. Conclusions

Reported cases of GBS manifesting after receiving COVID-19 vaccines across the world have been gathered and analyzed in this work. Mostly, cases manifested after receiving AstraZeneca and Pfizer vaccines. Affected individuals usually presented following the first dose, and although mortality was rare, average mEGOS was rather low (4 ± 3.57). Several variants of GBS were observed, but the BFP variant of GBS was of higher incidence than expected. Survival analysis resulted in a steep increase in complication risk if GBS manifests after 13 days of vaccination. Physicians should keep in mind the potential risk of GBS after vaccination, as urgent and quick management can help resolve the disease with minimal damage. Overall, the incidence seems rather sporadic, yet further investigation and assessment of causes and outcomes of this disease is needed, and future research focusing on predisposing factors are of important value.

CRedit authorship contribution statement

Mohammad Aladawi: Methodology, Project administration, Software, Visualization, Writing – original draft. **Hashem Abu Serhan:** Funding acquisition, Supervision, Validation, Writing – review & editing. **Elsayed Mohamed Hammad:** Data curation, Methodology, Project administration, Resources, Writing – original draft. **Ayman A. Mahmoud:** Conceptualization, Data curation, Formal analysis, Software. **Obaida Falah:** Data curation, Formal analysis, Resources, Software, Validation. **Ibrahim T. Abuawwad:** Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft. **Ahmad J. Taha:** Data curation, Formal analysis, Investigation, Project administration, Software. **Yasmeena Abdelall Kozaa:** Data curation, Formal analysis, Investigation, Methodology. **Mohammad T. Abuawwad:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Mohammad J. J. Taha:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft.

Declaration of Competing Interest

The authors have no conflicts of interest to declare.

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Statement of Ethics

An ethics statement is not applicable because this study is based exclusively on published literature.

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References

- [1] J.J. Sejvar, A.L. Baughman, M. Wise, O.W. Morgan, Population incidence of Guillain-Barré syndrome: a systematic review and meta-analysis, *Neuroepidemiology* 36 (2) (2011) 123–133.
- [2] A. McGrogan, G.C. Madle, H.E. Seaman, C.S. de Vries, The Epidemiology of Guillain-Barré Syndrome Worldwide, *Neuroepidemiology* 32 (2) (2009) 150–163.
- [3] J. Finsterer, Triggers of Guillain-Barré Syndrome: *Campylobacter jejuni* Predominates, *Int J. Mol. Sci.* 23 (22) (2022).
- [4] C. Vellozzi, S. Iqbal, K. Broder, Guillain-Barre syndrome, influenza, and influenza vaccination: the epidemiologic evidence, *Clin. Infect. Dis.* 58 (8) (2014) 1149–1155.
- [5] H. Fujimura, Chapter 21 - The Guillain-Barré syndrome, *Handbook of Clinical Neurology*, 115, Elsevier, 2013, pp. 383–402G. Said, C. Krarup.
- [6] Y. Hao, W. Wang, B.C. Jacobs, B. Qiao, M. Chen, D. Liu, et al., Antecedent infections in Guillain-Barré syndrome: a single-center, prospective study, *Ann. Clin. Transl. Neurol.* 6 (12) (2019) 2510–2517.
- [7] N. Shahrizaila, H.C. Lehmann, S. Kuwabara, Guillain-Barré syndrome, *Lancet* 397 (10280) (2021) 1214–1228.
- [8] H. Koike, Y. Fukami, R. Nishi, Y. Kawagashira, M. Iijima, M. Katsuno, et al., Ultrastructural mechanisms of macrophage-induced demyelination in Guillain-Barré syndrome, *J. Neurol. Neurosurg. Psychiatry* 91 (6) (2020) 650–659.
- [9] J.W. Griffin, C.Y. Li, T.W. Ho, P. Xue, C. Macko, C.Y. Gao, et al., Guillain-Barré syndrome in northern China. The spectrum of neuropathological changes in clinically defined cases, *Brain* 118 (Pt 3) (1995) 577–595.
- [10] M. Aladawi, M. Elfil, B. Abu-Esbeh, D. Abu Jazar, A. Armouti, A. Bayoumi, et al., Guillain Barre Syndrome as a Complication of COVID-19: A Systematic Review, *Can. J. Neurol. Sci.* 49 (1) (2022) 38–48.
- [11] A. Liberati, D.G. Altman, J. Tetzlaff, C. Mulrow, P.C. Gøtzsche, J.P. Ioannidis, et al., The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration, *PLoS Med* 6 (7) (2009) e1000100.
- [12] NIH. Study Quality Assessment Tools | NHLBI, NIH.
- [13] C. Fokke, B. van den Berg, J. Drenth, C. Walgaard, P.A. van Doorn, B.C. Jacobs, Diagnosis of Guillain-Barré syndrome and validation of Brighton criteria, *Brain* 137 (Pt 1) (2014) 33–43.
- [14] Modified Erasmus GBS Outcome Score (EGOS) at day 7 of admission.
- [15] A. Abicic, I. Adamec, M. Habek, Miller Fisher syndrome following Pfizer COVID-19 vaccine, *Neurol. Sci.* 43 (3) (2022) 1495–1497.
- [16] A. Abo-Zed, A. Pinevich, Guillain-Barré Syndrome, or Acute on Chronic Inflammatory Demyelinating Polyneuropathy, Following Moderna Covid-19 Vaccine, *Chest* 160 (4) (2021).
- [17] M. Aldeeb, L. Okar, S.S. Mahmud, G.A. Adeli, Could Guillain-Barre syndrome be triggered by COVID-19 vaccination? *Clin. Case Rep.* 10 (1) (2022) e05237.
- [18] K. Alekseyev, L. Didenko, B. Chaudhry, Guillain-Barre Syndrome Two Weeks after 1st Dose of Covid-19 Vaccine, *J. Phys. Med. Rehabil. Stud. Rep. SRC/JPMRS/166* 4 (1) (2022) 2.
- [19] D. Aljamea, Z. Al Sadah, A. Alokley, H. Altaweel, M. Alabdali, M. AlQarni, Neuro-immunological complications post COVID-19 vaccination: two case reports, *F1000Research* 11 (2022).
- [20] F. Alkolfat, A. Elfatraty, J. Mekky, A.S. Aly, Combined peripheral and central nervous system demyelination post-COVID-19 vaccination: A case report, *Neuroimmunol. Rep.* 2 (2022).
- [21] C.M. Allen, S. Ramsamy, A.W. Tarr, P.J. Tighe, W.L. Irving, R. Tanasescu, et al., Guillain-Barre Syndrome Variant Occurring after SARS-CoV-2 Vaccination, *Ann. Neurol.* 90 (2) (2021) 315–318.
- [22] A.S. Aly, F. Alkolfat, E.R. Mansour, S. Salama, Guillain-Barre syndrome following COVID-19 vaccination: a case report and an updated review, *Neuroimmunol. Rep.* 2 (2022).
- [23] V. Andreozzi, B. D'Arco, P. Pagliano, A. Toriello, P. Barone, Bilateral facial palsy after COVID-19 vaccination, *Neurol. Sci.* 43 (7) (2022) 4069–4079.
- [24] Z. Anjum, C. Iyer, S. Naz, V. Jaiswal, G. Nepal, M. Laguio-Vila, et al., Guillain-Barre syndrome after mRNA-1273 (Moderna) COVID-19 vaccination: A case report, *Clin. Case Rep.* 10 (4) (2022) e05733.
- [25] L. Arce Gálvez, L.A. Ramírez Abadía, C.A. de los Reyes Guevara, J.F. Hernández Orozco, Guillain-Barre syndrome after vaccination for Covid-19. The first report in Latin America, *Neurol. Perspect.* 1 (4) (2021) 236–238.
- [26] S. Azam, A. Khalil, A. Taha, Guillain-Barré Syndrome in a 67-year-old Male Post COVID-19 Vaccination (Astra Zeneca), *Am. J. Med. Case Rep.* 9 (8) (2021) 424–427.
- [27] F. Bax, G.L. Gigli, E. Belgrado, L. Brunelli, M. Valente, Guillain-Barre syndrome following Covid-19 immunization: a report of two cases, *Acta Neurol. Belg.* 122 (5) (2022) 1365–1367.
- [28] H. Bazrafshan, L.S. Mohamadi Jahromi, R. Parvin, A. Ashraf, A case of Guillain-Barre syndrome after the second dose of AstraZeneca COVID-19 vaccination, *Turk. J. Phys. Med Rehabil.* 68 (2) (2022) 295–299.
- [29] W. Bello Marq, A. AguirreIII, Guillain-Barré Syndrome Post-COVID 19 Vaccination with Sinovac CoronaVac in a 59-Year-Old Male: First Reported Case in the Philippines 6 (8 J. Infect. Dis. Epidemiol.2022.

- [30] M. Bellucci, F. Germano, S. Grisanti, C. Castellano, F. Tazza, E.M. Mobilia, et al., Case Report: Post-COVID-19 Vaccine Recurrence of Guillain-Barré Syndrome Following an Antecedent Parainfectious COVID-19-Related GBS, *Front. Immunol.* 13 (2022).
- [31] S. Benlamkadam, Y. Zakaria, L. Aballa, M. Chraa, N. Louhab, N. Kissani, Guillain-Barré syndrome post COVID-19 vaccine: Two cases report study, *J. Neurol. Sci.* 429 (2021).
- [32] A. Biswas, S.K. Pandey, D. Kumar, H. Vardhan, Post coronavirus disease-2019 vaccination Guillain-Barre syndrome, *Indian J. Public Health* 65 (4) (2021) 422–424.
- [33] G.B. Bonifacio, D. Patel, S. Cook, E. Purcaru, M. Couzins, J. Domjan, et al., Bilateral facial weakness with paraesthesia variant of Guillain-Barre syndrome following Vaxzevria COVID-19 vaccine, *J. Neurol. Neurosurg. Psychiatry* 93 (3) (2022) 341–342.
- [34] N. Bouattour, O. Hdiji, S. Sakka, E. Fakhfakh, K. Moalla, S. Daoud, et al., Guillain-Barre syndrome following the first dose of Pfizer-BioNTech COVID-19 vaccine: case report and review of reported cases, *Neurol. Sci.* 43 (2) (2022) 755–761.
- [35] O. Carranza, D. Babici, S. Waheed, F. Yousuf, Neurologic Sequela of COVID-19: Guillain-Barre Syndrome Following Johnson & Johnson COVID-19 Vaccination 4 14 *Cureus*, e24252.2022.
- [36] J.I. Castiglione, J.M. Crespo, L. Lecchini, F.O. Silveira, M.B. Luis, N. Cotti, et al., Bilateral facial palsy with paresthesias, variant of Guillain-Barre syndrome following COVID-19 vaccine: A case series of 9 patients, *Neuromuscul. Disord.* 32 (7) (2022) 572–574.
- [37] D. Censcak, L. Ungermann, I. Stetkarova, E. Ehler, Guillain-Barre Syndrome after First Vaccination Dose against COVID-19: Case Report, *Acta Med. (Hrad.-. Kral.)* 64 (3) (2021) 183–186.
- [38] M.C. Chang, Guillain-Barré syndrome with weakness confined to the bilateral upper extremities after ChAdOx1-S/ nCoV-19 vaccination, *Neurol. Asia* 27 (2) (2022) 497–502.
- [39] B. Chen, S. Lopez, E. Eggenberger, Miller Fisher syndrome after Pfizer BioNTech vaccine booster responsive to intravenous Ig treatment, *BMJ Case Rep.* 15 (11) (2022) e251361.
- [40] J.Y. Chun, S. Park, J. Jung, S.-H. Kim, T.-S. Kim, Y.J. Choi, et al., Guillain-Barré syndrome after vaccination against COVID-19, *Lancet Neurol.* 21 (2) (2022) 117–119.
- [41] G.F. da Silva, C.F. da Silva, R. Oliveira, F. Romancini, R.M. Mendes, A. Locks, et al., Guillain-Barre syndrome after coronavirus disease 2019 vaccine: A temporal association, *Clin. Exp. Neuroimmunol.* 13 (2) (2022) 92–94.
- [42] V. Dalwadi, D. Hancock, A.A. Ballout, A. Geraci, Axonal-Variant Guillain-Barre Syndrome Temporally Associated With mRNA-Based Moderna SARS-CoV-2 Vaccine, *Cureus* 13 (9) (2021) e18291.
- [43] Y.L. Dang, A. Bryson, Miller-Fisher Syndrome and Guillain-Barre Syndrome overlap syndrome in a patient post Oxford-AstraZeneca SARS-CoV-2 vaccination, *BMJ Case Rep.* 14 (11) (2021).
- [44] Z. Demissie, E. Tesfaye, A. Bitew, A. Worku, T. Haile, The First Ethiopian Case Report of Post-COVID-19 Vaccine Associated Guillain-Barre Syndrome, *Enormous J. Med. Sci. Curr. Res.* 2 (4) (2022) 01–05.
- [45] L. Donaldson, E. Margolin, Variant Guillain-Barre Syndrome Following SARS-CoV-2 Vaccination: Case Report and Review of the Literature, *Can. J. Neurol. Sci. / J. Can. Des. Sci. Neurol.* 3 (1) (2022).
- [46] M.S. Fakhari, L. Poorsaadat, B. Mahmoodiyeh, Guillain-Barre syndrome following COVID-19 vaccine: A case report, *Clin. Case Rep.* 10 (10) (2022) e6451.
- [47] J. Fernandes, S. Jaggermath, V. Rammarnine, S.R. Mohammed, C. Khan, A. Panday, Neurological Conditions Following COVID-19 Vaccinations: Chance or Association? *Cureus* (2022).
- [48] J. Finsterer, Guillain-Barre syndrome 15 days after COVID-19 despite SARS-CoV-2 vaccination, *IDCases* 25 (2021) e01226.
- [49] J. Finsterer, Exacerbating Guillain-Barre Syndrome Eight Days after Vector-Based COVID-19 Vaccination, *Case Rep. Infect. Dis.* 2021 (2021) 3619131.
- [50] T. Fukushima, M. Tomita, S. Ikeda, N. Hattori, A case of sensory ataxic Guillain-Barre syndrome with immunoglobulin G anti-GM1 antibodies following the first dose of mRNA COVID-19 vaccine BNT162b2 (Pfizer), *QJM* 115 (1) (2022) 25–27.
- [51] F. Germano, M. Bellucci, S. Grisanti, A. Beronio, M. Grazzini, E. Coco, et al., COVID-19 vaccine-related Guillain-Barre syndrome in the Liguria region of Italy: A multicenter case series, *J. Neurol. Sci.* 440 (2022) 120330.
- [52] P.N.G. Gonzales, S.G. Villaraza, A.M. Punzalan-Sotelo, Guillain-Barré Syndrome: A Case Report of Post Covid-19 Vaccination in the Philippines, *J. Health Med. Sci.* 5 (3) (2022).
- [53] P.Y. Gunawan, P. Tiffani, L. Lalisang, Guillain-Barre Syndrome Following SARS-CoV-2 Vaccination: A Case Report, *Clin. Psychopharmacol. Neurosci.* 20 (4) (2022) 777–780.
- [54] T. Hasan, M. Khan, F. Khan, G. Hamza, Case of Guillain-Barre syndrome following COVID-19 vaccine, *BMJ Case Rep.* 14 (6) (2021).
- [55] S. Hemalatha, A.G. Krishnan, T. Loganathan, J. Thirumaran, S. Blessy, J. Bera, et al., A Case Report on Post COVID -19 Vaccine Associated Guillain-Barré Syndrome, *J. Pharm. Res. Int.* (2021) 454–458.
- [56] A. Hiltz, A. Schreiber, A. Singh, A Clinical Case of COVID-19 Vaccine-Associated Guillain-Barre Syndrome, *Am. J. Case Rep.* 23 (2022) e936896.
- [57] A. Hiraga, M. Muto, S. Kuwabara, Loss of Taste as an Initial Symptom of a "Facial Diplegia and Paresthesia" Variant of Guillain-Barre Syndrome, *Intern Med* 61 (19) (2022) 2957–2959.
- [58] D.L. Hughes, J.A. Brunn, J. Jacobs, P.K. Todd, F.K. Askari, R.J. Fontana, Guillain-Barre Syndrome After COVID-19 mRNA Vaccination in a Liver Transplantation Recipient With Favorable Treatment Response, *Liver Transpl.* 28 (1) (2022) 134–137.
- [59] U. Ilyas, Z. Umar, R. Bhargal, D. Shah, B. Fayman, Guillain-Barre Syndrome: A Sequela of the Original COVID-19 Infection or Vaccination, *Cureus* 14 (8) (2022) e28044.
- [60] A. Introna, F. Caputo, C. Santoro, T. Guerra, M. Ucci, D.M. Mezzapesa, et al., Guillain-Barre syndrome after AstraZeneca COVID-19-vaccination: A causal or casual association? *Clin. Neurol. Neurosurg.* 208 (2021) 106887.
- [61] E. Jain, K. Pandav, P. Regmi, G. Michel, I. Altschuler, Facial Diplegia: A Rare, Atypical Variant of Guillain-Barre Syndrome and Ad26.COV2.S Vaccine, *Cureus* 13 (7) (2021) e16612.
- [62] R.S. Jain, A. Vyas, P.V. Sripadma, K. Rao, Post COVID-19 Vaccination GBS- Association or Causation? *Ann. Indian Acad. Neurol.* 25 (2) (2022) 294–296.
- [63] G. Kanabar, P. Wilkinson, Guillain-Barre syndrome presenting with facial diplegia following COVID-19 vaccination in two patients, *BMJ Case Rep.* 14 (10) (2021).
- [64] N. Karimi, R. Boostani, F. Fatehi, A. Panahi, A. Asghar Okhovat, B. Ziaadini, et al., Guillain-Barre Syndrome and COVID-19 Vaccine: A Report of Nine Patients, *Basic Clin. Neurosci.* 12 (5) (2021) 703–710.
- [65] B. Khadka, K. Khanal, Post COVID-19 vaccine Guillain-Barre syndrome, *J. Nepal Health Res. Coun.* 19 (4) (2022) 852–854.
- [66] S. Khanal, S.B. Khanal, Guillain-Barre Syndrome Following Ad26. Cov2. S Covid-19 Vaccine: A Case Report, *J. Nobel Med. Coll.* 10 (2) (2021) 63–64.
- [67] J.E. Kim, Y.G. Min, J.Y. Shin, Y.N. Kwon, J.S. Bae, J.J. Sung, et al., Guillain-Barre Syndrome and Variants Following COVID-19 Vaccination: Report of 13 Cases, *Front Neurol.* 12 (2021) 820723.
- [68] J.W. Kim, Y.G. Kim, Y.C. Park, S. Choi, S. Lee, H.J. Min, et al., Guillain-Barre Syndrome After Two COVID-19 Vaccinations: Two Case Reports With Follow-up Electrodiagnostic Study, *J. Korean Med. Sci.* 37 (7) (2022) e58.
- [69] N. Kim, J.H. Kim, J.S. Park, Guillain-Barre syndrome associated with BNT162b2 COVID vaccination: a first case report from South Korea, *Neurol. Sci.* 43 (3) (2022) 1491–1493.
- [70] Y. Kim, Z. Zhu, P. Kochar, P. Gavigan, D. Kaur, A. Kumar, A Pediatric Case of Sensory Predominant Guillain-Barre Syndrome Following COVID-19 Vaccination, *Child Neurol. Open* 9 (2022) 2329048×221074549.
- [71] R. Koren, J.C. Do, The Development of Guillain Barre Syndrome Subsequent to Administration of Ad26.COV2.S Vaccine, *J. Neurol. Res. Rev. Rep.* 1–3 (2021).
- [72] Y. Kripalani, V. Lakkappan, L. Parulekar, A. Shaikh, R. Singh, P. Vyas, A Rare Case of Guillain-Barre Syndrome following COVID-19 Vaccination, *Eur. J. Case Rep. Intern Med* 8 (9) (2021) 002707.
- [73] P. Kulsrichawaroj, O. Sanmaneechai, O. Wittawatmongkol, K. Chokephaibulkit, Polyneuritis Cranialis Associated with BNT162b2 mRNA COVID-19 Vaccine in a Healthy Adolescent, *Vaccin. (Basel)* 10 (1) (2022).
- [74] T.A. Lanman, C. Wu, H. Cheung, N. Goyal, M. Greene, Guillain-Barre Syndrome with Rapid Onset and Autonomic Dysfunction Following First Dose of Pfizer-BioNTech COVID-19 Vaccine: A Case Report, *Neurohospitalist* 12 (2) (2022) 388–390.
- [75] S. Lee, S.-i Oh, Guillain-Barré Syndrome Following the ChAdOx1-S/nCoV-19 Vaccine: A Case Report, *J. Korean Assoc. EMG-Electrodiagn. Med.* 24 (1) (2022) 7–10.
- [76] W. Leemans, S. Antonis, W. De Vooght, R. Lemmens, P. Van Damme, Neuromuscular complications after COVID-19 vaccination: a series of eight patients, *Acta Neurol. Belg.* 122 (3) (2022) 753–761.
- [77] H. Liang, Y. Cao, W. Zhong, Z. Ma, J. Liu, H. Chen, Miller-Fisher syndrome and Guillain-Barre syndrome overlap syndrome following inactivated COVID-19 vaccine: Case report and scope review, *Hum. Vaccin. Immunother.* 18 (6) (2022) 2125753.
- [78] L. Ling, S.M. Bagshaw, P.M. Villeneuve, Guillain-Barre syndrome after SARS-CoV-2 vaccination in a patient with previous vaccine-associated Guillain-Barre syndrome, *CMAJ* 193 (46) (2021) E1766–E1769.
- [79] P. Luitel, B. Poudel, D. Upadhyay, S. Paudel, N. Tiwari, B.P. Gajurel, et al., Guillain-Barre syndrome following coronavirus disease vaccine: First report from Nepal, *SAGE Open Med Case Rep.* 10 (2022) 2050313×221100876.
- [80] E. Malamud, S.I. Otallah, J.B. Caress, D.J. Lapid, Guillain-Barre Syndrome After COVID-19 Vaccination in an Adolescent, *Pedia Neurol.* 126 (2022) 9–10.
- [81] B.V. Maramattom, P. Krishnan, R. Paul, S. Padmanabhan, S. Cherukudal Vishnu Nampoothiri, A.A. Syed, et al., Guillain-Barre Syndrome following ChAdOx1-S/ nCoV-19 Vaccine, *Ann. Neurol.* 90 (2) (2021) 312–314.
- [82] F.G. Masuccio, C. Comi, C. Solaro, Guillain-Barre syndrome following COVID-19 vaccine mRNA-1273: a case report, *Acta Neurol. Belg.* 122 (5) (2022) 1369–1371.
- [83] A.S. Matarneh, A.H. Al-Battah, K. Farooqui, M. Ghamoodi, M. Alhatou, COVID-19 vaccine causing Guillain-Barre syndrome, a rare potential side effect, *Clin. Case Rep.* 9 (9) (2021) e04756.
- [84] N. McKean, C. Chirco, Guillain-Barre syndrome after COVID-19 vaccination, *BMJ Case Rep.* 14 (7) (2021).
- [85] Y.G. Min, W. Ju, Y.E. Ha, J.J. Ban, S.A. Lee, J.J. Sung, et al., Sensory Guillain-Barre syndrome following the ChAdOx1 nCoV-19 vaccine: Report of two cases and review of literature, *J. Neuroimmunol.* 359 (2021) 577691.
- [86] Z.P. Morehouse, A. Paulus, S.A. Jasti, X. Bing, A Rare Variant of Guillain-Barre Syndrome Following Ad26.COV2.S Vaccination, *Cureus* 13 (9) (2021) e18153.
- [87] K. Mosna, P. Vadkerti, L. Papp, M. Palkovic, P. Janega, P. Babal, Guillain-Barré Syndrome with Lethal Outcome Following COVID-19 Vaccination - Case Report Supported by Autopsy Examination, *Open Neurol. J.* 16 (1) (2022).
- [88] H. Mozdehpanah, S. Mavandadi, S. Paybast, Guillain-Barré Syndrome After Receiving the First Dose of Oxford-AstraZeneca SARS-CoV-2 Vaccine, *Acta Med. Iran.* 60 (6) (2022).
- [89] S. Nagalli, N. Shankar Kikkeri, Sub-acute Onset of Guillain-Barre Syndrome Post-mRNA-1273 Vaccination: a Case Report, *SN Compr. Clin. Med* 4 (1) (2022) 41.

- [90] G. Nagdev, G. Chavan, G. Sahu, P.D. Devasilpa Raju, COVID-19 Vaccination a Cause of Guillain-Barré Syndrome? A Case Series, *Cureus* (2022).
- [91] K. Nanatsue, M. Takahashi, S. Itaya, K. Abe, A. Inaba, A case of Miller Fisher syndrome with delayed onset peripheral facial nerve palsy after COVID-19 vaccination: a case report, *BMC Neurol.* 22 (1) (2022) 309.
- [92] A. Nasiri, Q. Haddad, N. Alazwary, I. Ghazzi, M. Elkarouri, Guillain-Barre Syndrome Post COVID-19 Vaccination: Case Report, *Int. J. Infect. Dis. Ther.* 7 (1) (2022).
- [93] N.A. Nasuelli, F. De Marchi, M. Cecchin, I. De Paoli, S. Onorato, R. Pettinaroli, et al., A case of acute demyelinating polyradiculoneuropathy with bilateral facial palsy after ChAdOx1 nCoV-19 vaccine, *Neurol. Sci.* 42 (11) (2021) 4747–4749.
- [94] Y. Nishiguchi, H. Matsuyama, K. Maeda, A. Shindo, H. Tomimoto, Miller Fisher syndrome following BNT162b2 mRNA coronavirus 2019 vaccination, *BMC Neurol.* 21 (1) (2021) 452.
- [95] S. Ogata, Y. Ishii, K. Asano, E. Kobayashi, S. Kubota, K. Takahashi, et al., Sensory Ataxic Guillain-Barre Syndrome with Dysgeusia after mRNA COVID-19 Vaccination, *Intern Med* 61 (11) (2022) 1757–1760.
- [96] O. Ogbonor, H. Seth, Z. Min, N. Bhanot, Guillain-Barre syndrome following the first dose of SARS-CoV-2 vaccine: A temporal occurrence, not a causal association, *IDCases* 24 (2021) e01143.
- [97] S.U. Patel, R. Khurram, A. Lakhani, B. Quirk, Guillain-Barre syndrome following the first dose of the chimpanzee adenovirus-vectored COVID-19 vaccine, *ChAdOx1, BMJ Case Rep.* 14 (4) (2021).
- [98] F.J.C. Pirola, B.A.M. Santos, G.F. Sapienza, L.Y. Cetrangolo, C.H.W.G. Geranutti, P.H.P. de Aguiar, Miller-Fisher syndrome after first dose of Oxford/AstraZeneca coronavirus disease 2019 vaccine: a case report, *J. Med. Case Rep.* 16 (1) (2022) 437.
- [99] M.B. Prado, K.J.B. Adiao, Facial Diplegia as the Sole Manifestation of Post-Vaccination Guillain-Barre Syndrome: A Case Report and Literature Review, *Neurohospitalist* 12 (3) (2022) 508–511.
- [100] A. Prasad, G. Hurlburt, S. Podury, M. Tandon, S. Kingree, S. Sriwastava, A Novel Case of Bilateral Diplegia Variant of Guillain-Barre Syndrome Following Janssen COVID-19 Vaccination, *Neurol. Int* 13 (3) (2021) 404–409.
- [101] S.J. Rao, S. Khurana, G. Murthy, E.T. Dawson, N. Jazebi, C.J. Haas, A case of Guillain-Barre syndrome following Pfizer COVID-19 vaccine, *J. Community Hosp. Intern Med Perspect.* 11 (5) (2021) 597–600.
- [102] A. Razok, A. Shams, A. Almeer, M. Zahid, Post-COVID-19 vaccine Guillain-Barre syndrome; first reported case from Qatar, *Ann. Med Surg. (Lond.)* 67 (2021) 102540.
- [103] J. Richardson-May, E. Purcaru, C. Campbell, C. Hillier, B. Parkin, Guillain-Barré Syndrome and Unilateral Optic Neuritis Following Vaccination for COVID-19: A Case Report and Literature Review *Neuro-Ophthalmol.* 2022, , 1–7..
- [104] Rohilla R., Kakkar A.K., Divyashree K., Mohindra R., Suri V. Recombinant protein subunit COVID-19 vaccine-induced Guillain-Barre Syndrome in an adolescent: A case report. *Br J Clin Pharmacol.* 2022.
- [105] A. Rossetti, G. Ghehman, M. O'Hare, J.M. Kosowsky, Guillain-Barre Syndrome Presenting as Facial Diplegia after COVID-19 Vaccination: A Case Report, *J. Emerg. Med* 61 (6) (2021) e141–e145.
- [106] Saraswat B., Kumar S., Rawal D., Johri S., Rai P.K., Grover A. Acute Onset Quadriplegia Post COVID 19 Vaccination: A Case Report. 2021.
- [107] R. Scendoni, C. Petrelli, G. Scaloni, F.O. Logullo, Electromyoneurography and laboratory findings in a case of Guillain-Barre syndrome after second dose of Pfizer COVID-19 vaccine, *Hum. Vaccin Immunother.* 17 (11) (2021) 4093–4096.
- [108] A.R. Siddiqi, T. Khan, M.J. Tahir, M.S. Asghar, M.S. Islam, Z. Yousaf, Miller Fisher syndrome after COVID-19 vaccination: Case report and review of literature, *Med. (Baltim.)* 101 (20) (2022) e29333.
- [109] O. Sosa-Hernández, S. Sánchez-Cardoza, Case report of Guillain-Barré Syndrome after COVID BNT162b2 mRNA vaccine, *Vacunas (Engl. Ed.)* 23 (2022) 67–69.
- [110] S.C. Su, R.K. Lyu, C.W. Chang, W.J. Tseng, The First Guillain-Barré Syndrome After SARS-CoV-2 Vaccination in Taiwan, *Acta Neurol. Taiwan* 31 (1) (2022) 46–51.
- [111] S. Tabatabaee, F. Rezanian, S.M.J. Alwedaie, E. Malekdar, Z. Badi, S. M. Tabatabaee, et al., Post COVID-19 vaccination Guillain-Barre syndrome: three cases, *Hum. Vaccin Immunother.* 18 (5) (2022) 2045153.
- [112] K. Takahashi, Y. Tomoda, S. Kadena, T. Kanbayashi, S. Kobayashi, R. Kato, Guillain-Barre syndrome after BNT162b2 (Pfizer-BioNTec) vaccination, *QJM* 115 (5) (2022) 331–333.
- [113] H.L. Thant, R. Morgan, M.M. Paese, T. Persaud, J. Diaz, L. Hurtado, Guillain-Barre Syndrome After Ad26.COV2.S Vaccination, *Am. J. Case Rep.* 23 (2022) e935275.
- [114] M. Trimboli, P. Zoleo, G. Arabia, A. Gambardella, Guillain-Barre syndrome following BNT162b2 COVID-19 vaccine, *Neurol. Sci.* 42 (11) (2021) 4401–4402.
- [115] S. Waheed, A. Bayas, F. Hindi, Z. Rizvi, P.S. Espinosa, Neurological Complications of COVID-19: Guillain-Barre Syndrome Following Pfizer COVID-19 Vaccine, *Cureus* 13 (2) (2021) e13426.
- [116] M. Yamakawa, K. Nakahara, T. Nakanishi, T. Nomura, M. Ueda, Miller Fisher Syndrome Following Vaccination against SARS-CoV-2, *Intern Med* 61 (7) (2022) 1067–1069.
- [117] A.S. Zubair, J.Y. Bae, K. Desai, Facial Diplegia Variant of Guillain-Barre Syndrome in Pregnancy Following COVID-19 Vaccination: A Case Report, *Cureus* 14 (2) (2022) e22341.
- [118] Mathieu E., Ritchie H., Rodés-Guirao L., Appel C., Giattino C., Hasel J., et al. Coronavirus Pandemic (COVID-19). *Our World in Data.* 2020.
- [119] L. Schirmer, V. Worthington, U. Solloch, V. Loleit, V. Grummel, N. Lakdawala, et al., Higher frequencies of HLA DQB1*05:01 and anti-glycosphingolipid antibodies in a cluster of severe Guillain-Barré syndrome, *J. Neurol.* 263 (10) (2016) 2105–2113.
- [120] H. Yoshikawa, [Epidemiology of Guillain-Barré Syndrome], *Brain Nerve* 67 (11) (2015) 1305–1311.
- [121] WHO. Statement of the WHO Global Advisory Committee on Vaccine Safety (GACVS) COVID-19 subcommittee on reports of Guillain-Barré Syndrome (GBS) following adenovirus vector COVID-19 vaccines. 2021.
- [122] A. Lleo, P.M. Battezzati, C. Selmi, M.E. Gershwin, M. Podda, Is autoimmunity a matter of sex? *Autoimmun. Rev.* 7 (8) (2008) 626–630.
- [123] S. Frisancho-Kiss, S.E. Davis, J.F. Nyland, J.A. Frisancho, D. Cihakova, M. A. Barrett, et al., Cutting edge: cross-regulation by TLR4 and T cell Ig mucin-3 determines sex differences in inflammatory heart disease, *J. Immunol.* 178 (11) (2007) 6710–6714.
- [124] P.A. McCombe, J.M. Greer, I.R. Mackay, Sexual dimorphism in autoimmune disease, *Curr. Mol. Med* 9 (9) (2009) 1058–1079.
- [125] P.Q. Yin, Y.Y. Sun, H.P. Chen, G.Z. Li, D. Zhong, Genome-wide gene expression analysis of peripheral leukocytes in relation to the male predominance of Guillain-Barre syndrome: differential gene expression between male and female patients, *Int J. Neurosci.* 126 (6) (2016) 531–541.
- [126] P.A. McCombe, T.A. Hardy, R.J. Nona, J.M. Greer, Sex differences in Guillain Barré syndrome, chronic inflammatory demyelinating polyradiculoneuropathy and experimental autoimmune neuritis, *Front Immunol.* 13 (2022) 1038411.
- [127] H. Abu Serhan, A. Abdelaal, M.T. Abuawwad, M.J.J. Taha, S. Irshaidat, L. Abu Serhan, et al., Ocular Vascular Events following COVID-19 Vaccines: A Systematic Review, *Vaccin. [Internet]* 10 (12) (2022).
- [128] Data OWI. Share of people with at least one dose of COVID-19 vaccine by age. *Our World in Data.* 2023.
- [129] K.E. Hanson, K. Goddard, N. Lewis, B. Fireman, T.R. Myers, N. Bakshi, et al., Incidence of Guillain-Barré Syndrome After COVID-19 Vaccination in the Vaccine Safety Datalink, *JAMA Netw. Open* 5 (4) (2022) e228879.
- [130] B. van den Berg, C. Walgaard, J. Drenthen, C. Fokke, B.C. Jacobs, P.A. van Doorn, Guillain-Barré syndrome: pathogenesis, diagnosis, treatment and prognosis, *Nat. Rev. Neurol.* 10 (8) (2014) 469–482.
- [131] S. Kim, H.J. Han, H.Y. Shin, S.W. Kim, Old age and multiple comorbidity are associated with delayed diagnosis of Guillain-Barré syndrome, *Sci. Rep.* 12 (1) (2022) 9913.
- [132] S. Peric, I. Berisavac, O. Stojiljkovic Tamas, S. Rajic, M. Babic, M. Cvijanovic, et al., Guillain-Barré syndrome in the elderly, *J. Peripher. Nerv. Syst.* 21 (2) (2016) 105–110.
- [133] P. Kumar, R. Charaniya, A. Bahl, A. Ghosh, J. Dixit, Facial Diplegia with Paresthesia: An Uncommon Variant of Guillain-Barre Syndrome, *J. Clin. Diagn. Res* 10 (7) (2016) Od01, 2.
- [134] C.M. Allen, S. Ramsamy, A.W. Tarr, P.J. Tighe, W.L. Irving, R. Tanasescu, et al., Guillain-Barré Syndrome Variant Occurring after SARS-CoV-2 Vaccination, *Ann. Neurol.* 90 (2) (2021) 315–318.
- [135] G.B. Bonifacio, D. Patel, S. Cook, E. Purcaru, M. Couzins, J. Domjan, et al., Bilateral facial weakness with paraesthesia variant of Guillain-Barré syndrome following Vaxzevria COVID-19 vaccine, *J. Neurol. Neurosurg. Psychiatry* 93 (3) (2022) 341–342.
- [136] F.X. Heinz, K. Stiasny, Distinguishing features of current COVID-19 vaccines: knowns and unknowns of antigen presentation and modes of action, *NPJ Vaccin.* 6 (1) (2021) 104.
- [137] A. Hilt, A. Schreiber, A. Singh, A Clinical Case of COVID-19 Vaccine-Associated Guillain-Barré Syndrome, *Am. J. Case Rep.* 23 (2022) e936896.
- [138] H. Koike, A. Chiba, M. Katsuno, Emerging Infection, Vaccination, and Guillain-Barré Syndrome: A Review, *Neurol. Ther.* 10 (2) (2021) 523–537.
- [139] C. Tard, C.A. Maurage, A.M. de Paula, F. Cassim, A. Delval, G. Kuchcinski, et al., Anti-pan-neurofascin IgM in COVID-19-related Guillain-Barré syndrome: Evidence for a nodo-paranodopathy, *Neurophysiol. Clin.* 50 (5) (2020) 397–399.
- [140] S.K.K. Chua, Q.Y. Soh, S.E. Saffari, E.K. Tan, Prognosis of Guillain-Barré Syndrome Linked to COVID-19 Vaccination, *Brain Sci.* 12 (6) (2022).
- [141] E.M.A. COVID-19 Vaccine Janssen: Guillain-Barré syndrome listed as a very rare side effect. *European Medicines Agency.* 2021.
- [142] H.L. Thant, R. Morgan, M.M. Paese, T. Persaud, J. Diaz, L. Hurtado, Guillain-Barré Syndrome After Ad26.COV2.S Vaccination, *Am. J. Case Rep.* 23 (2022) e935275.
- [143] K.T. Mills, A. Stefanescu, J. He, The global epidemiology of hypertension, *Nat. Rev. Nephrol.* 16 (4) (2020) 223–237.
- [144] P. Wen, L. Wang, H. Liu, L. Gong, H. Ji, H. Wu, et al., Risk factors for the severity of Guillain-Barré syndrome and predictors of short-term prognosis of severe Guillain-Barré syndrome, *Sci. Rep.* 11 (1) (2021) 11578.
- [145] I. González-Suárez, I. Sanz-Gallego, F.J. Rodríguez de Rivera, J. Arpa, Guillain-Barré syndrome: natural history and prognostic factors: a retrospective review of 106 cases, *BMC Neurol.* 13 (2013) 95.
- [146] H. Ghazanfar, R. Qazi, A. Ghazanfar, S. Iftikhar, Significance of Brighton Criteria in the Early Diagnosis and Management of Guillain-Barré Syndrome, *Cureus* 12 (5) (2020) e8318.
- [147] S. Roy, N. Jain, E. Narang, J. Singh, N. Gupta, Impact of COVID Lockdown on the Presentation of Nasal Foreign Bodies to a Tertiary Care Center, *Ear Nose Throat J.* 101 (10) (2022) Np453–np8.
- [148] G. Del Vecchio Blanco, E. Calabrese, L. Biancone, G. Monteleone, O.A. Paoluzi, The impact of COVID-19 pandemic in the colorectal cancer prevention, *Int J. Colorectal Dis.* 35 (10) (2020) 1951–1954.
- [149] M.L. Mansueto, G. Zagni, C. Sartori, B.A. Olivares Bermudez, B. Righi, C. Catellani, et al., Late diagnosis of severe long-standing autoimmune hypothyroidism after the first lockdown for the Covid-19 pandemic: clinical features and follow-up, *Acta Biomed.* 92 (S1) (2022) e2021239.
- [150] H. Brüssow, COVID-19: vaccination problems, *Environ. Microbiol.* 23 (6) (2021) 2878–2890.